Atmega 328p-Au

U1
ATMEGA328P-AU

D3	1	(PCINT19/OC2B/INT1)PD3	PD2(INT0/PCINT18)	32	D2
D4	2	(PCINT20/XCK/T0)PD4	PD1(TXD/PCINT17)	31	TX0
	3	GND	PD0(RXD/PCINT16)	30	RX0
	4	VCC	PC6(RESET/PCINT14)	29	Reset
	5	GND	PC5(ADC5/SCL/PCINT13)	28	A5_SCL
	6	VCC	PC4(ADC4/SDA/PCINT12)	27	A4_SDA
	7	(PCINT6/XTAL1/TOSC1)PB6	PC3(ADC3/PCINT11)	26	A3
	8	(PCINT7/XTAL2/TOSC2)PB7	PC2(ADC2/PCINT10)	25	A2
D5_PWM	9	(PCINT21/OC0B/T1)PD5	PC1(ADC1/PCINT9)	24	A1
D6_PWM	10	(PCINT22/OC0A/AIN0)PD6	PC0(ADC0/PCINT8)	23	A0
D7	11	(PCINT23/AIN1)PD7	ADC7	22	A7
D8	12	(PCINT0/CLKO/ICP1)PB0	GND	21	
D9_PWM	13	(PCINT1/OC1A)PB1	AREF	20	
D10_PWM	14	(PCINT2/SS/OC1B)PB2	ADC6	19	A6
D11	15	(PCINT3/OC2A/MOSI)PB3	AVCC	18	
D12	16	(PCINT4/MISO)PB4	PB5(SCK/PCINT5)	17	D13

CSTNE16M0V53Z000R0 X1
100nF C1
R1 10k
C2 100nF C3 10µF

LORA_E32

J1 LoraModule
Befestigung J17
Lora_M0
Lora_M1
Lora_RX
Lora_TX
Lora_AUX
+3.3V
GND

ANZEIGE LEDS

RV_LED1 1k
RV_LED2 1k
kopplungLED D30
D31 PWR_LED
Jumper_LED
J15

Reset Taster

Reset1
Reset
C4 100nF

Test Points

TP5 tpGND
TP3 tp_+5V
TP4 tp_+3V3
TP6 +12V tp_+12V

OPVs Impedanzwandler

Solar+ 470k R2
R3 470k
LM358 U11A
LM358 U11B
BAT_Volt
A6_Solar
A6_BAT_Volt
U11C LM358
C5 100nF

DS3231 RTC

DS3231_RTC J3
A5_SCL
A4_SDA

Analog Pin Multiplexer

U8
CD4052B
A6_Solar
A6
A6_BAT_Volt
Sens_Vent7
Sens_Vent6
A7
Sens_Vent5
Sens_Vent8
D4
D5_PWM
C6 100nF

SIM7080G

PWR 2Sec auf Low in setup um modul zu booten
SIM_RX
SIM_TX
SIM_PWR
Sim7000
J5

I/O Expander mit Ventilerweiterung

MCP23S17-E/SS U12
R4 4.7k
R5 4.7k
R6 10k
R7 10k
R8 10k
A5_SCL
A4_SDA
Vent1_D2
Vent1_D1
Vent2_D2
Vent2_D1
Vent3_D2
Vent3_D1
Vent4_D2
Vent4_D1
J4
Erweiterung_Ventile1
J20
IO1_VentErw
Sens_Vent7
Sens_Vent6
Sens_Vent5
Sens_Vent8

PINOUT SPANNUNGSVERSORGUNG SOLAR

Battery1
BH-18650-A1BJ021
Bat-
Bat+
tp_Bat+
TP1
SIM_BAT
B2B-PH-K-S(LF)(SN)
Solar-
Solar+
tp_Solar+
TP2
J8
Solar1
J7
USB_IN_Bat_volt
BAT_Volt
USB_Volt
Bat_Solar_Vout

Magnetventile

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DCDCs

File: DCDCs.kicad_sch

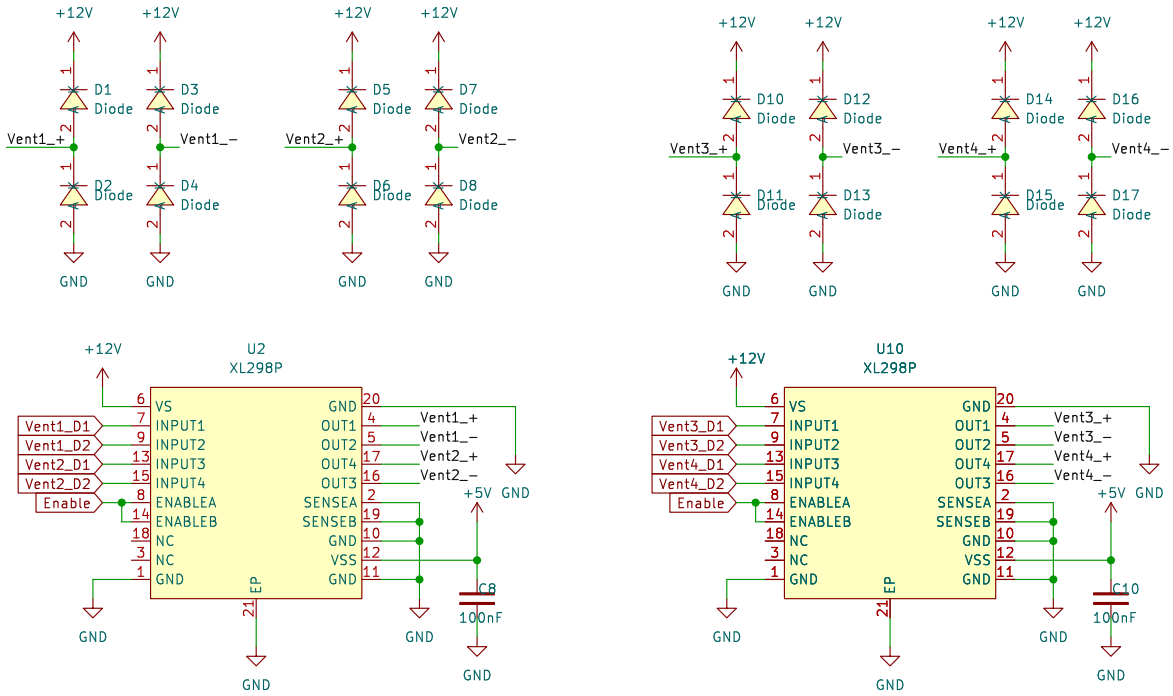
Bootloader/Code Upload

Gegenstück
J2
Reset
D13
D12
RX0
D11
TX0
Bootload

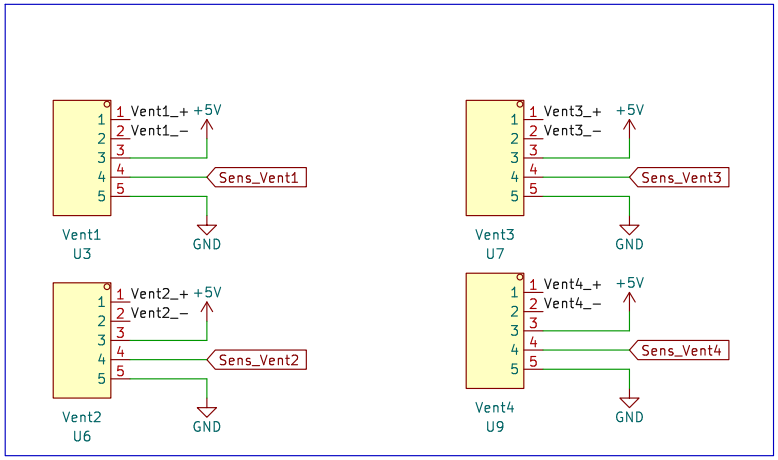
Connections

D11	SIM_RX	A3	Sens_Vent4
D12	SIM_TX	A2	Sens_Vent3
D13	SIM_PWR	A1	Sens_Vent2
		A0	Sens_Vent1
IO1_VentErw	D2	D10_PWM	Lora_AUX
Enable	D3	D9_PWM	Lora_M0
		D8	Lora_M1
		D7	Lora_RX
		D6_PWM	Lora_TX

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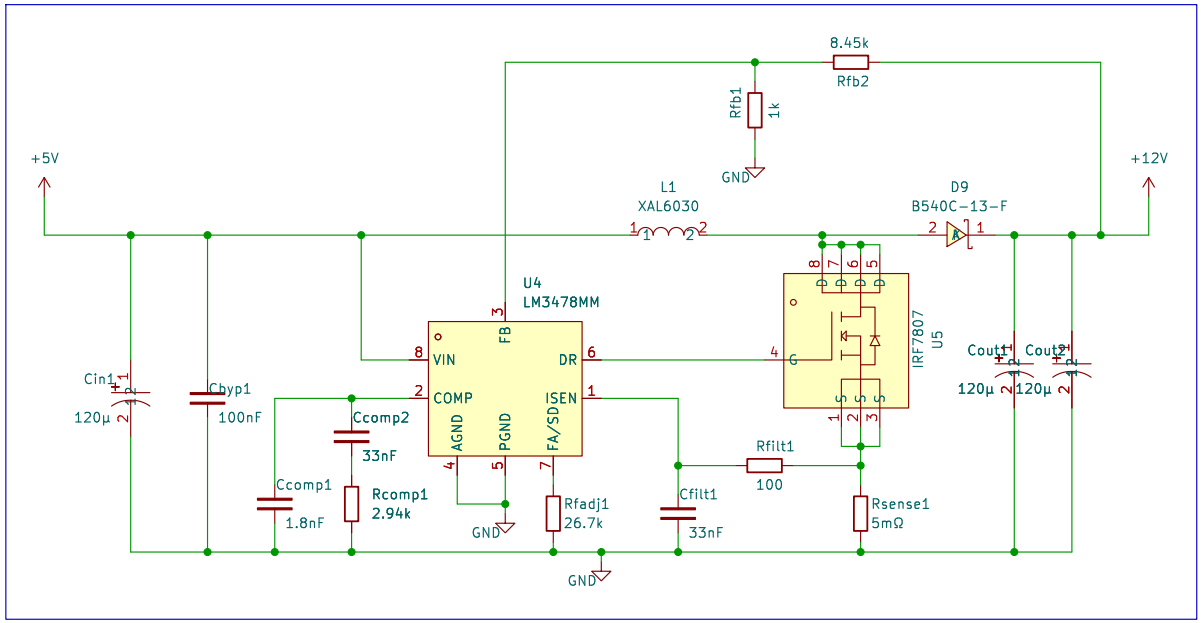


Stecker Sensoren & Ventile



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5 to 12Volt DCDC



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Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad 7.0.10		Id: 3/3